

Initiating Search

November 10, 2025, 10:01 PM

• Search:

CAS Lexicon Search:

Skraup reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (86)	 References	View Results
Exported: Retrieved Related Reaction Results (32)	 Reactions	View Results

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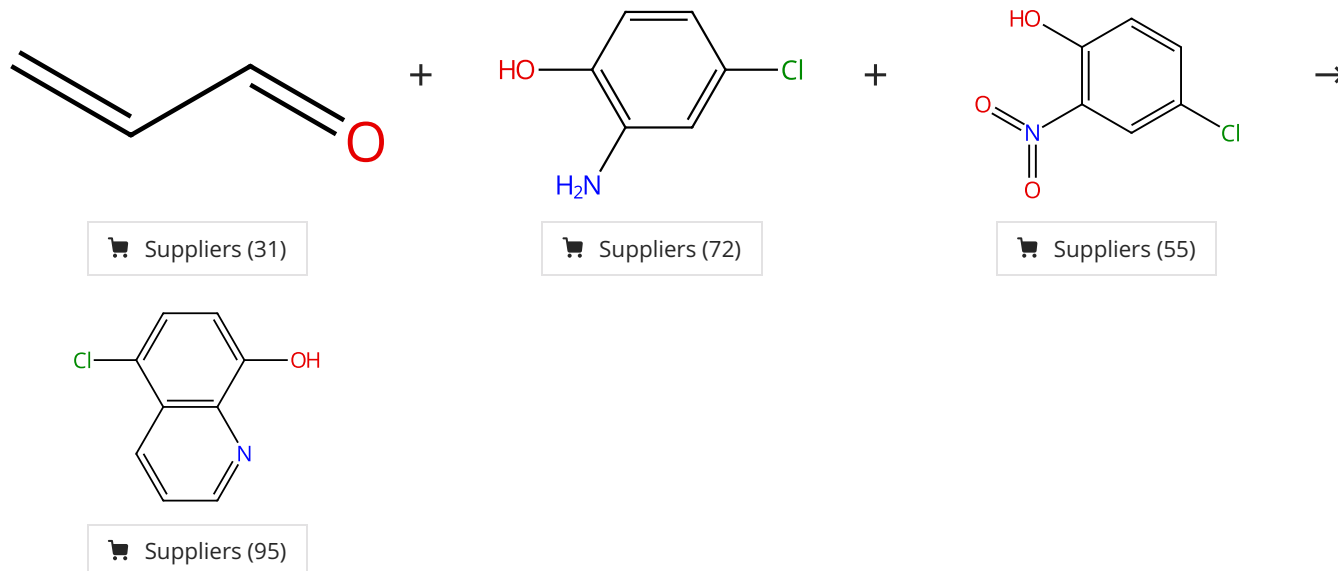
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Reactions (32)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-260-CAS-21149475

Steps: 1 Yield: 100%

Method for preparing 5-chloro-8-hydroxyquinoline using acrolein

- 1.1 Reagents: Acetic acid, Hydrochloric acid
Solvents: Water; 5 h, 90 °C; 1 h, reflux
- 1.2 Reagents: Sodium hydroxide; pH 7, rt

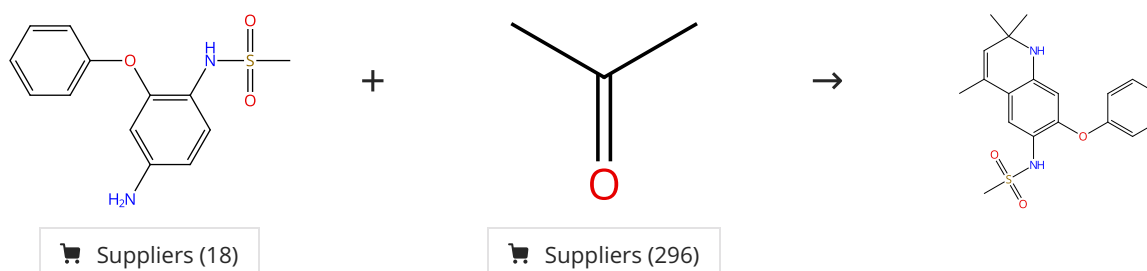
Assignee: Lier Chemical Co., Ltd.

China, CN108191753 A 2018-06-22

[PatentPak](#) available

Scheme 2 (1 Reaction)

Steps: 1 Yield: 80%



31-260-CAS-11203604

Steps: 1 Yield: 80%

Efficient catalyst of ceric ammonium nitrate for one-pot synthesis of 2,2,4-trimethyl-1,2-dihydroquinolines

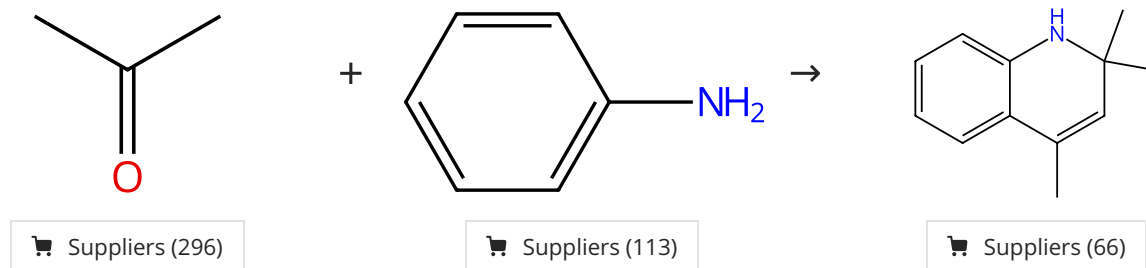
- 1.1 Catalysts: Ceric ammonium nitrate; 24 h, 55 °C

By: Durgadas, Shylaprasad; et al

Letters in Organic Chemistry (2010), 7(4), 306-310.

Scheme 3 (2 Reactions)

Steps: 1 Yield: 80%



31-260-CAS-979425

Steps: 1 Yield: 80%

Efficient catalyst of ceric ammonium nitrate for one-pot synthesis of 2,2,4-trimethyl-1,2-dihydroquinolines

By: Durgadas, Shylaprasad; et al

Letters in Organic Chemistry (2010), 7(4), 306-310.

1.1 Catalysts: Ceric ammonium nitrate; 24 h, 50 - 55 °C

31-260-CAS-18312999

Steps: 1

Synthesis and Biological Evaluation of Cyclopentaquinoline Derivatives as Nonsteroidal Glucocorticoid Receptor Antagonists

By: Eda, Masahiro; et al

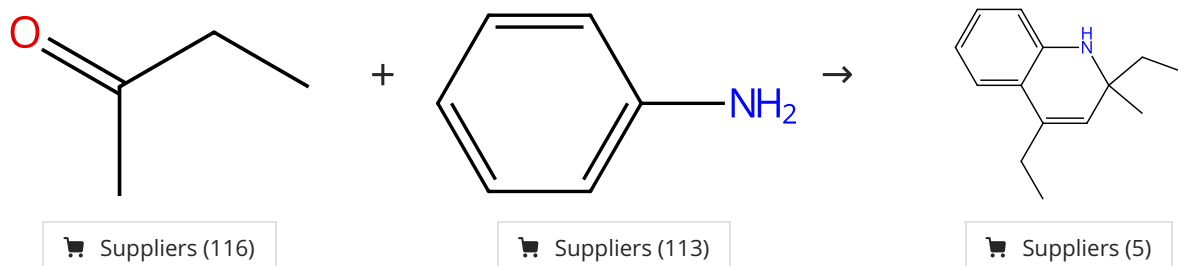
Journal of Medicinal Chemistry (2015), 58(12), 4918-4926.

1.1 Catalysts: Scandium triflate
Solvents: Acetone; rt

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 78%



31-260-CAS-2499812

Steps: 1 Yield: 78%

Efficient catalyst of ceric ammonium nitrate for one-pot synthesis of 2,2,4-trimethyl-1,2-dihydroquinolines

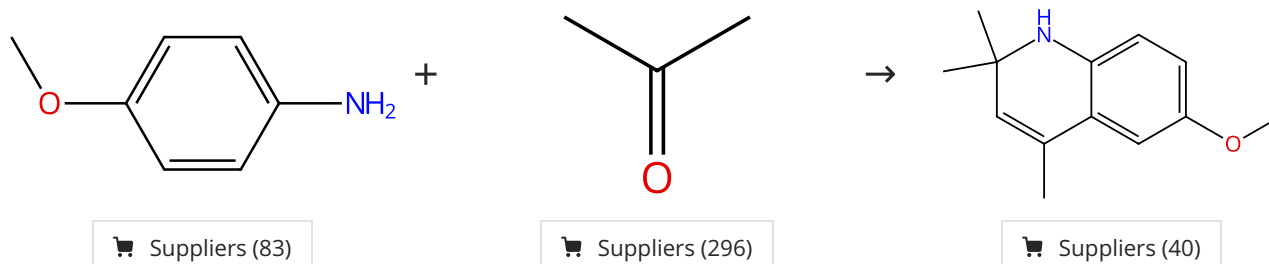
By: Durgadas, Shylaprasad; et al

Letters in Organic Chemistry (2010), 7(4), 306-310.

1.1 Catalysts: Ceric ammonium nitrate; 24 h, 50 - 55 °C

Scheme 5 (1 Reaction)

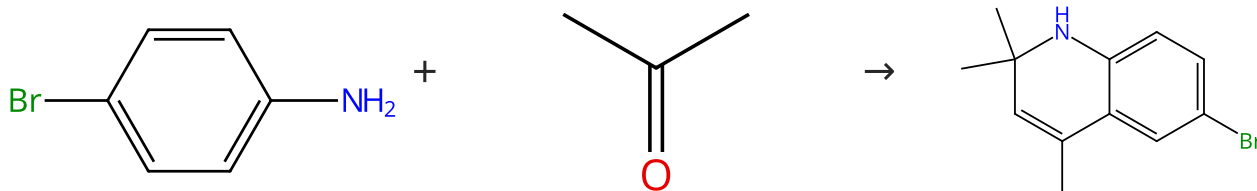
Steps: 1 Yield: 75%



31-260-CAS-7459990	Steps: 1 Yield: 75%	Efficient catalyst of ceric ammonium nitrate for one-pot synthesis of 2,2,4-trimethyl-1,2-dihydroquinolines
1.1 Catalysts: Ceric ammonium nitrate; 24 h, 50 - 55 °C		By: Durgadas, Shylaprasad; et al Letters in Organic Chemistry (2010), 7(4), 306-310.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 75%



Chemical reaction scheme showing the synthesis of 2,2,4-trimethyl-1,2-dihydroquinoline-5-bromide from 4-bromoaniline and acetone.

4-bromoaniline (Br, NH₂) + Acetone (O) → 2,2,4-trimethyl-1,2-dihydroquinoline-5-bromide (Br, NH)

 Suppliers (83)

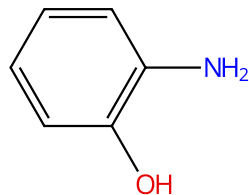
 Suppliers (296)

 Suppliers (38)

31-260-CAS-13749782	Steps: 1 Yield: 75%	Efficient catalyst of ceric ammonium nitrate for one-pot synthesis of 2,2,4-trimethyl-1,2-dihydroquinolines
1.1 Catalysts: Ceric ammonium nitrate; 24 h, 50 - 55 °C		By: Durgadas, Shylaprasad; et al Letters in Organic Chemistry (2010), 7(4), 306-310.

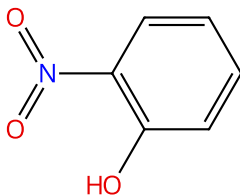
Scheme 7 (1 Reaction)

Steps: 1 Yield: 72%



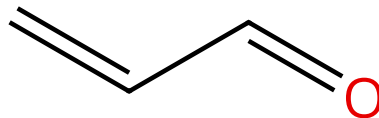
 Suppliers (85)

+



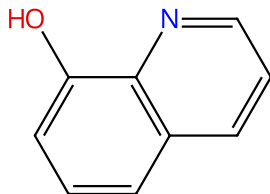
 Suppliers (94)

+



 Suppliers (31)

→

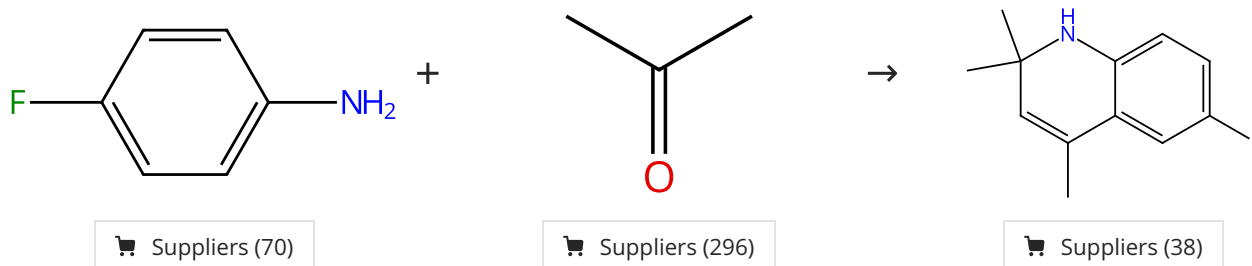


 Suppliers (136)

31-260-CAS-14844940	Steps: 1 Yield: 72%	Synthesis of 8-hydroxyquinoline by improved Skraup method
1.1 Reagents: Acetic acid, Hydrochloric acid Solvents: Water; rt → 100 °C		By: Liu, Duan; et al
1.2 3 h, 95 - 100 °C; 1 h, 95 - 100 °C		Huaxue Yu Shengwu Gongcheng (2009), 26(8), 43-44, 47.
1.3 Reagents: Sodium hydroxide Solvents: Water; pH 7, cooled		

Scheme 8 (1 Reaction)

Steps: 1 Yield: 72%



31-260-CAS-9493670

Steps: 1 Yield: 72%

Efficient catalyst of ceric ammonium nitrate for one-pot synthesis of 2,2,4-trimethyl-1,2-dihydroquinolines

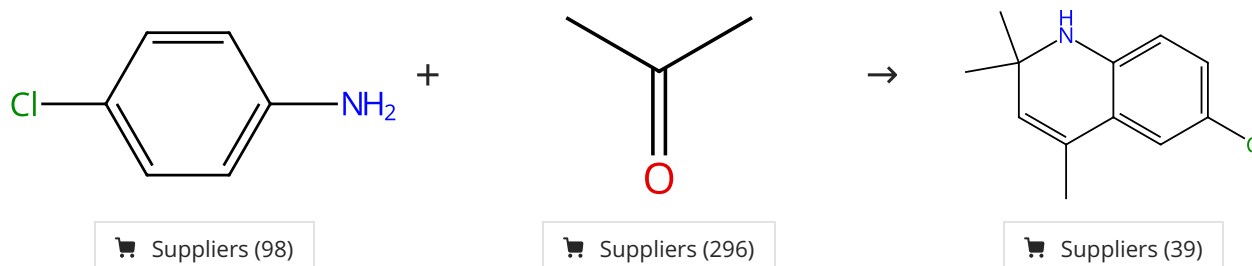
1.1 Catalysts: Ceric ammonium nitrate; 24 h, 50 - 55 °C

By: Durgadas, Shylaprasad; et al

Letters in Organic Chemistry (2010), 7(4), 306-310.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 70%



31-260-CAS-11619564

Steps: 1 Yield: 70%

Efficient catalyst of ceric ammonium nitrate for one-pot synthesis of 2,2,4-trimethyl-1,2-dihydroquinolines

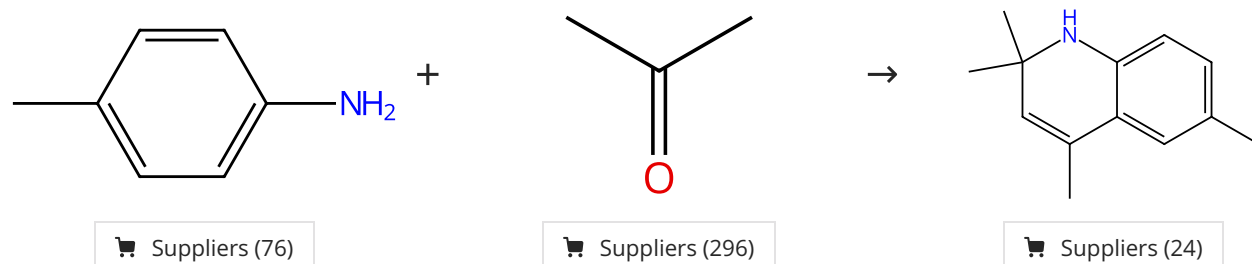
1.1 Catalysts: Ceric ammonium nitrate; 24 h, 50 - 55 °C

By: Durgadas, Shylaprasad; et al

Letters in Organic Chemistry (2010), 7(4), 306-310.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 70%



31-260-CAS-5207598

Steps: 1 Yield: 70%

Efficient catalyst of ceric ammonium nitrate for one-pot synthesis of 2,2,4-trimethyl-1,2-dihydroquinolines

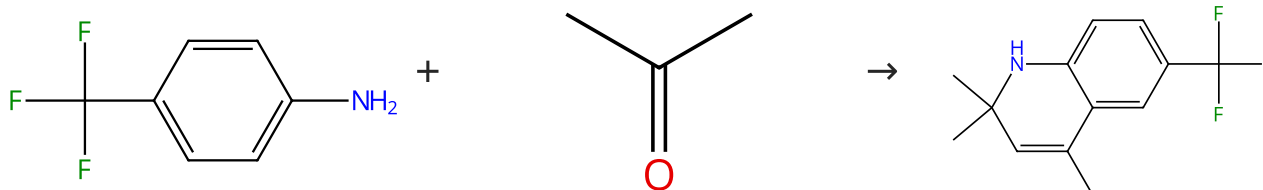
1.1 Catalysts: Ceric ammonium nitrate; 24 h, 50 - 55 °C

By: Durgadas, Shylaprasad; et al

Letters in Organic Chemistry (2010), 7(4), 306-310.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 68%



Suppliers (86)

Suppliers (296)

Suppliers (3)

31-260-CAS-332287

Steps: 1 Yield: 68%

Efficient catalyst of ceric ammonium nitrate for one-pot synthesis of 2,2,4-trimethyl-1,2-dihydroquinolines

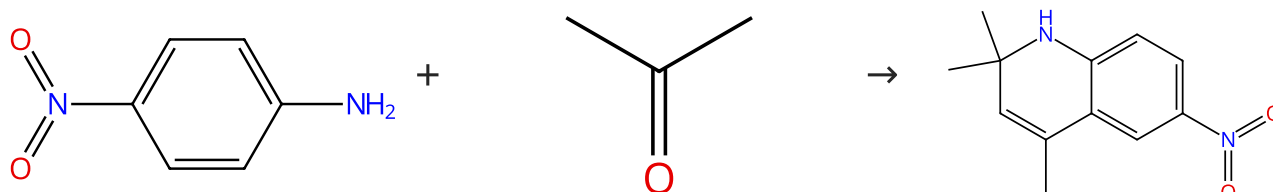
By: Durgadas, Shylaprasad; et al

Letters in Organic Chemistry (2010), 7(4), 306-310.

1.1 Catalysts: Ceric ammonium nitrate; 24 h, 50 - 55 °C

Scheme 12 (1 Reaction)

Steps: 1 Yield: 65%



Suppliers (93)

Suppliers (296)

Supplier (1)

31-260-CAS-3114187

Steps: 1 Yield: 65%

Efficient catalyst of ceric ammonium nitrate for one-pot synthesis of 2,2,4-trimethyl-1,2-dihydroquinolines

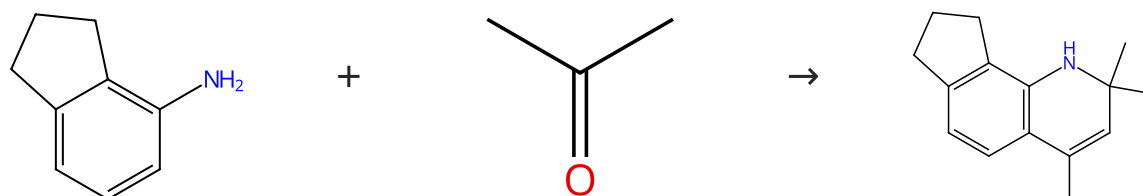
By: Durgadas, Shylaprasad; et al

Letters in Organic Chemistry (2010), 7(4), 306-310.

1.1 Catalysts: Ceric ammonium nitrate; 24 h, 50 - 55 °C

Scheme 13 (1 Reaction)

Steps: 1 Yield: 60%



Suppliers (73)

Suppliers (296)

Suppliers (2)

31-260-CAS-18051377

Steps: 1 Yield: 60%

Synthesis and Biological Evaluation of Cyclopenta quinoline Derivatives as Nonsteroidal Glucocorticoid Receptor Antagonists

By: Eda, Masahiro; et al

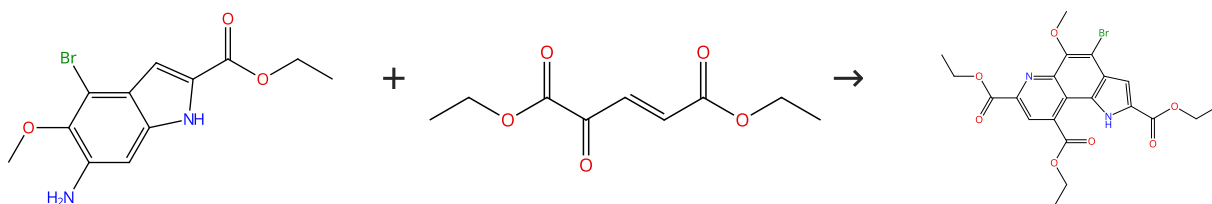
Journal of Medicinal Chemistry (2015), 58(12), 4918-4926.

1.1 Catalysts: Scandium triflate
Solvents: Acetone; 7 d, rt

Experimental Protocols

Scheme 14 (2 Reactions)

Steps: 1 Yield: 59%



31-309-CAS-21402437

Steps: 1 Yield: 59%

Process for preparation of pyrroloquinolinequinone (PQQ) intermediates

1.1 Solvents: Dichloromethane; 24 h, rt

Assignee: Shandong Camasy Biotechnology Co. Ltd.

World Intellectual Property Organization, WO2018205299 A1 2018-11-15

[PatentPak](#) available

31-309-CAS-21229445

Steps: 1 Yield: 59%

Process for preparation of pyrroloquinolinequinone (PQQ) intermediates

1.1 Solvents: Dichloromethane; 24 h, rt

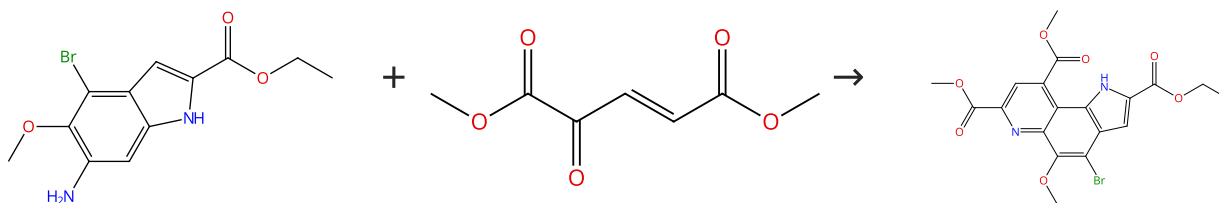
Assignee: Shandong Camasy Biotechnology Co., Ltd.

China, CN107089982 A 2017-08-25

[PatentPak](#) available

Scheme 15 (2 Reactions)

Steps: 1 Yield: 58%



Suppliers (32)

31-309-CAS-21400665

Steps: 1 Yield: 58%

Process for preparation of pyrroloquinolinequinone (PQQ) intermediates

1.1 Solvents: Dichloromethane; 24 h, rt

Assignee: Shandong Camasy Biotechnology Co. Ltd.

World Intellectual Property Organization, WO2018205299 A1 2018-11-15

[PatentPak](#) available

31-309-CAS-21230513

Steps: 1 Yield: 58%

Process for preparation of pyrroloquinolinequinone (PQQ) intermediates

1.1 Solvents: Dichloromethane; 24 h, rt

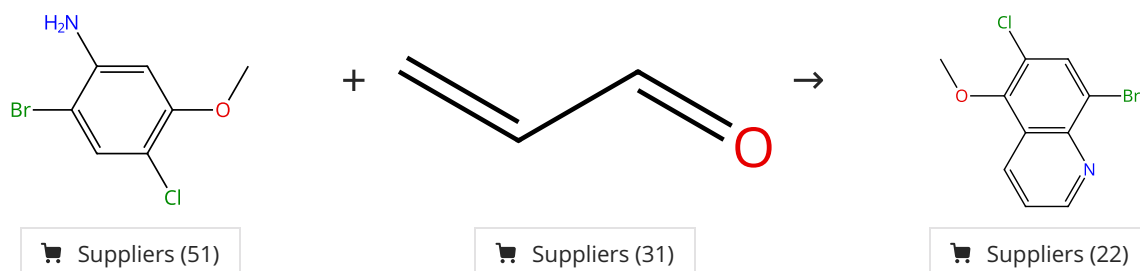
Assignee: Shandong Camasy Biotechnology Co., Ltd.

China, CN107089982 A 2017-08-25

[PatentPak](#) available

Scheme 16 (1 Reaction)

Steps: 1 Yield: 52%



31-260-CAS-14899395

Steps: 1 Yield: 52%

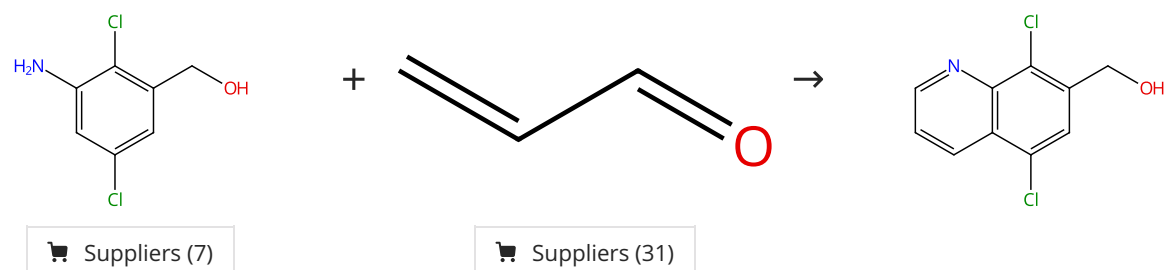
Synthesis of 5-hydroxyquinolines

1.1 Reagents: Hydrochloric acid
Solvents: Water

By: Li, Jianke; et al
Tetrahedron Letters (2010), 51(30), 3876-3878.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 48%



31-260-CAS-3456419

Steps: 1 Yield: 48%

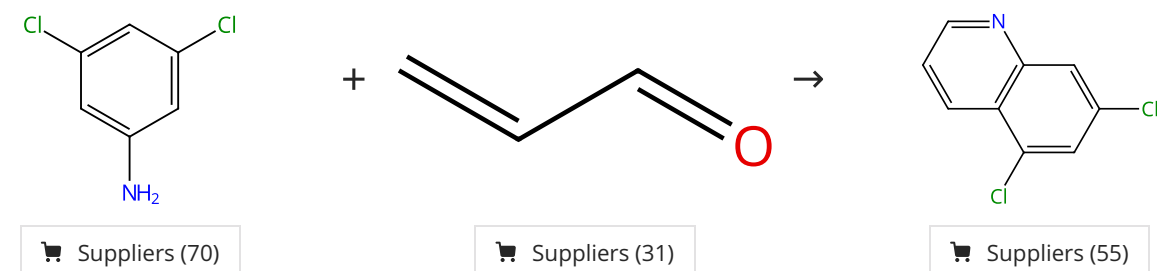
Synthesis of 5-hydroxyquinolines

1.1 Reagents: Hydrochloric acid
Solvents: Toluene, Water

By: Li, Jianke; et al
Tetrahedron Letters (2010), 51(30), 3876-3878.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 38%



31-260-CAS-5872146

Steps: 1 Yield: 38%

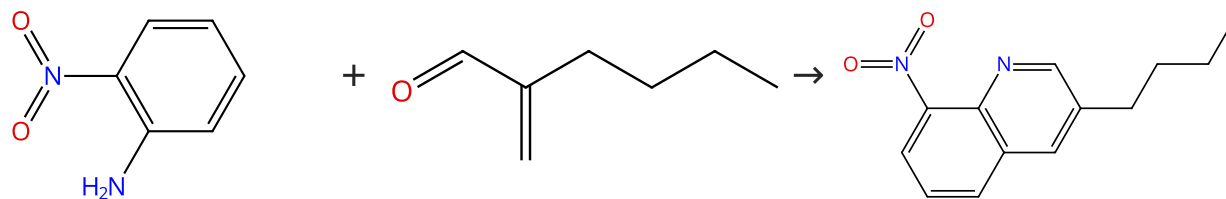
Synthesis of 5-hydroxyquinolines

1.1 Reagents: Hydrochloric acid
Solvents: Water

By: Li, Jianke; et al
Tetrahedron Letters (2010), 51(30), 3876-3878.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 35%



Suppliers (79)

Suppliers (22)

31-260-CAS-18191748

Steps: 1 Yield: 35%

A Systematic Study on the Synthesis of n-Butyl Substituted 8-Aminoquinolines

By: Koseoglu, Ahmet; et al

Journal of Heterocyclic Chemistry (2016), 53(1), 263-270.

1.1 Reagents: Arsenic oxide (As₂O₅), Sulfuric acid

Solvents: Water; rt → 100 °C

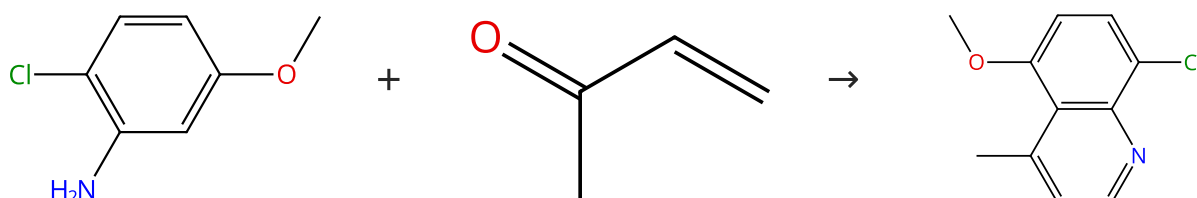
1.2 < 120 °C; overnight, 110 °C; cooled

1.3 Reagents: Sodium hydroxide

Solvents: Water; neutralized, cooled

Scheme 20 (1 Reaction)

Steps: 1 Yield: 35%



Suppliers (79)

Suppliers (25)

Suppliers (5)

31-260-CAS-6192850

Steps: 1 Yield: 35%

Synthesis of 5-hydroxyquinolines

By: Li, Jianke; et al

Tetrahedron Letters (2010), 51(30), 3876-3878.

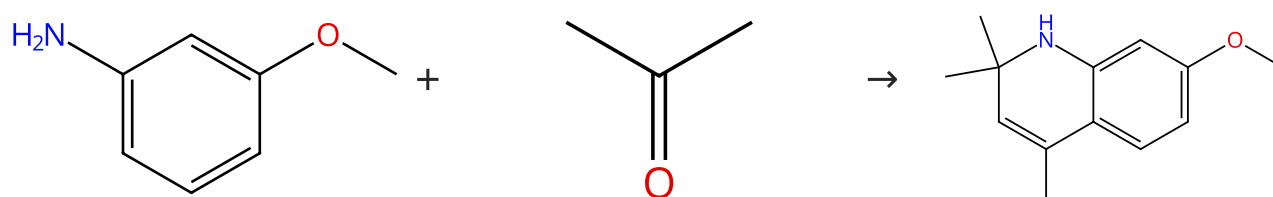
1.1 Reagents: Hydrochloric acid

Solvents: Toluene, Water; 1 h, reflux

1.2 Reagents: Zinc chloride; 30 min, rt; 1.5 h, 0 °C

Scheme 21 (1 Reaction)

Steps: 1 Yield: 35%



Suppliers (75)

Suppliers (296)

Suppliers (61)

31-260-CAS-18049282

Steps: 1 Yield: 35%

Synthesis and Biological Evaluation of Cyclopenta quinoline Derivatives as Nonsteroidal Glucocorticoid Receptor Antagonists

By: Eda, Masahiro; et al

Journal of Medicinal Chemistry (2015), 58(12), 4918-4926.

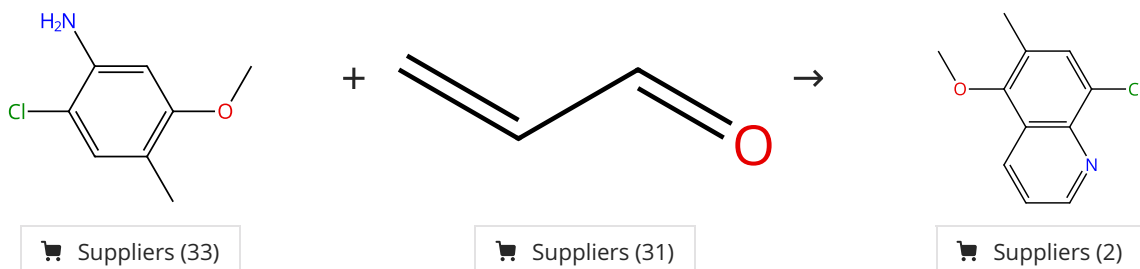
1.1 Reagents: Catechol, Iodine

Solvents: Acetone; 4 d, reflux

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 30%



31-260-CAS-8316534

Steps: 1 Yield: 30%

Synthesis of 5-hydroxyquinolines

1.1 Reagents: Hydrochloric acid
Solvents: Toluene, Water; 1 h, reflux

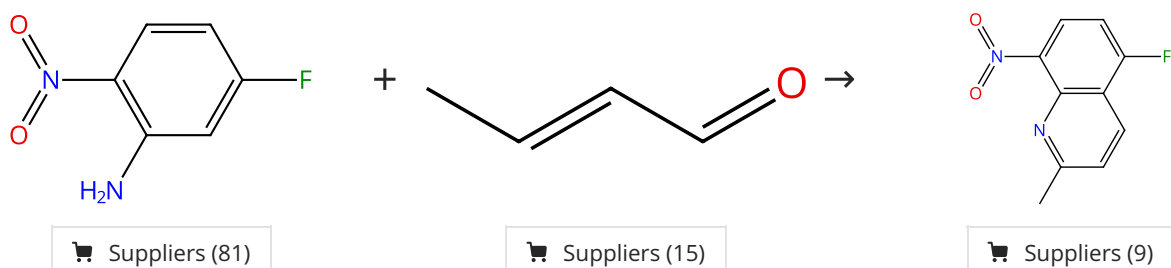
By: Li, Jianke; et al

1.2 Reagents: Zinc chloride; 30 min, rt; 1.5 h, 0 °C

Tetrahedron Letters (2010), 51(30), 3876-3878.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 28%



31-260-CAS-21172112

Steps: 1 Yield: 28%

Synthesis of fluorine-containing 1,10-phenanthrolines using mild versions of Skraup and Doebner-von Miller reactions

1.1 Reagents: Arsenic oxide (As₂O₅), Sulfuric acid
Solvents: Water; rt → 70 °C

By: Luedtke, Carsten; et al

1.2 3 h, 70 °C; 90 min, 70 °C → 120 °C; cooled

Journal of Fluorine Chemistry (2017), 193, 98-105.

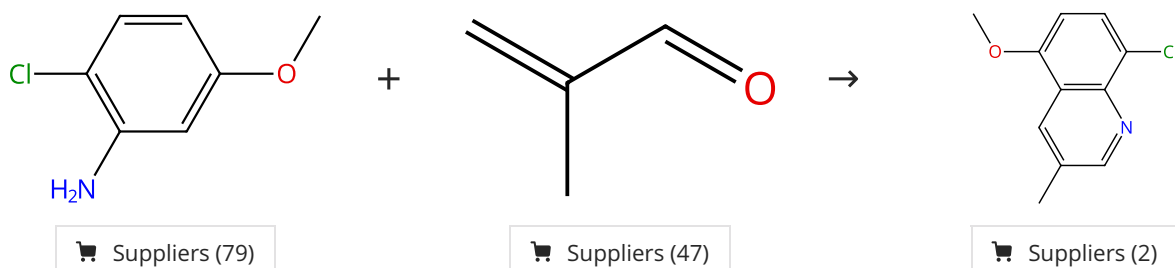
1.3 Solvents: Water; 0 °C

1.4 Reagents: Ammonia
Solvents: Water; pH 6, 0 °C

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 27%



31-260-CAS-4015037

Steps: 1 Yield: 27%

Synthesis of 5-hydroxyquinolines

1.1 Reagents: Hydrochloric acid
Solvents: Toluene, Water; 1 h, reflux

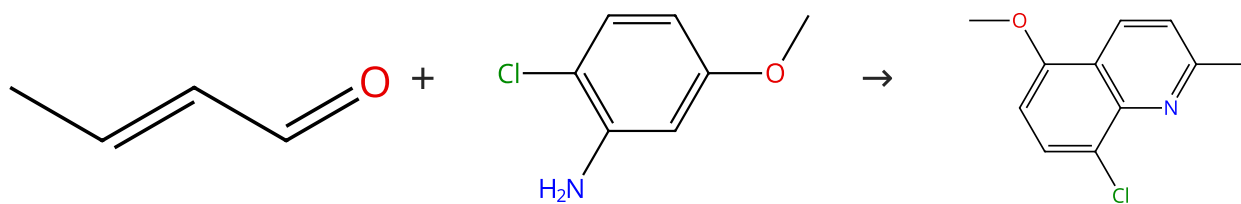
By: Li, Jianke; et al

1.2 Reagents: Zinc chloride; 30 min, rt; 1.5 h, 0 °C

Tetrahedron Letters (2010), 51(30), 3876-3878.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 23%



Suppliers (15)

Suppliers (79)

Suppliers (12)

31-260-CAS-1720705

Steps: 1 Yield: 23%

Synthesis of 5-hydroxyquinolines

1.1 Reagents: Hydrochloric acid
Solvents: Toluene, Water; 1 h, reflux

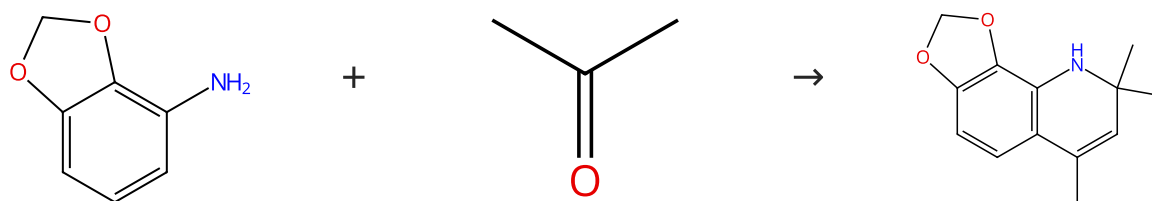
By: Li, Jianke; et al

1.2 Reagents: Zinc chloride; 30 min, rt; 1.5 h, 0 °C

Tetrahedron Letters (2010), 51(30), 3876-3878.

Scheme 26 (1 Reaction)

Steps: 1



Suppliers (60)

Suppliers (296)

Suppliers (3)

31-260-CAS-18046093

Steps: 1

Synthesis and Biological Evaluation of Cyclopenta quinoline Derivatives as Nonsteroidal Glucocorticoid Receptor Antagonists

1.1 Catalysts: Scandium triflate
Solvents: Acetone; rt

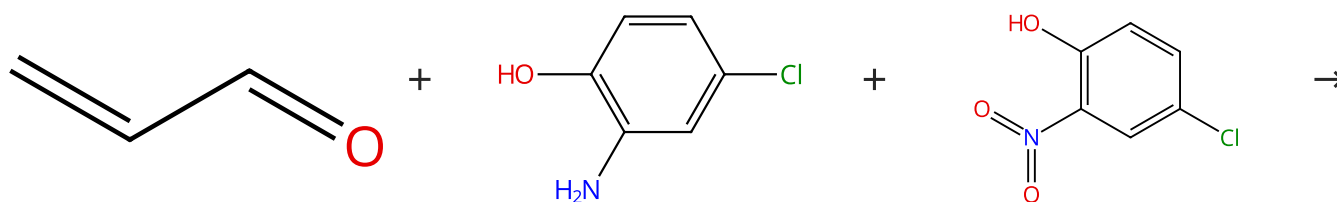
By: Eda, Masahiro; et al

Experimental Protocols

Journal of Medicinal Chemistry (2015), 58(12), 4918-4926.

Scheme 27 (1 Reaction)

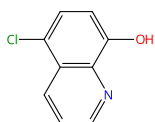
Steps: 1



Suppliers (31)

Suppliers (72)

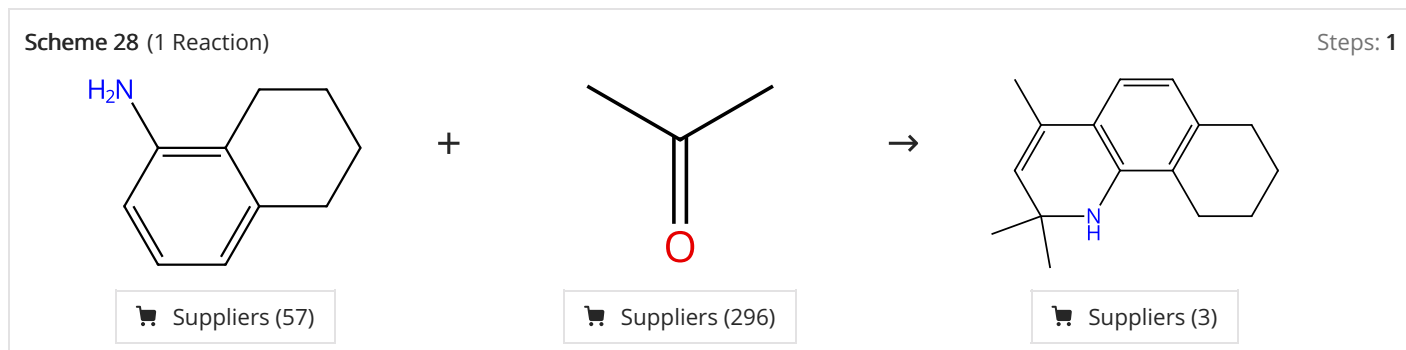
Suppliers (55)



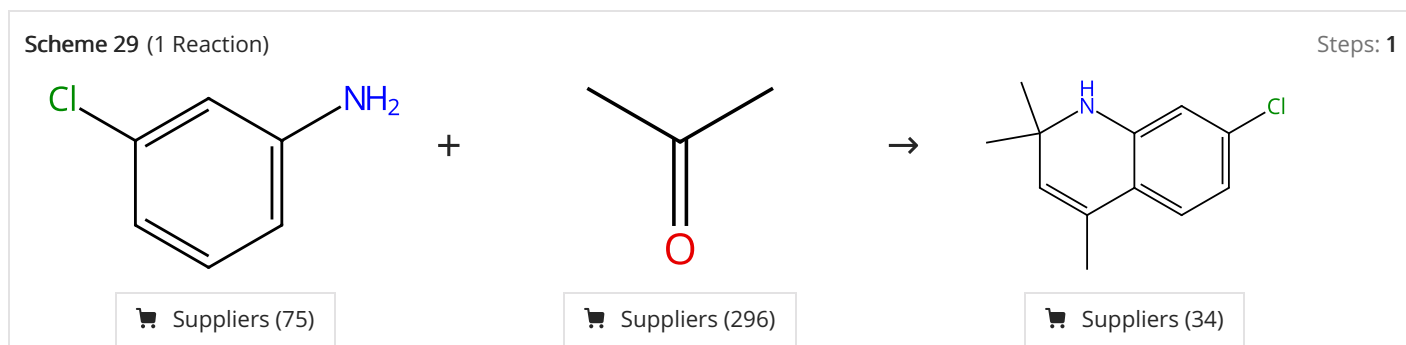
• HCl

Suppliers (9)

31-260-CAS-21149104	Steps: 1	Method for preparing 5-chloro-8-hydroxyquinoline using acrolein
1.1 Reagents: Acetic acid, Hydrochloric acid Solvents: Water; 5 h, 90 °C; 1 h, reflux		
		Assignee: Lier Chemical Co., Ltd.
		China, CN108191753 A 2018-06-22
		PatentPak available



31-260-CAS-18045765	Steps: 1	Synthesis and Biological Evaluation of Cyclopenta quinoline Derivatives as Nonsteroidal Glucocorticoid Receptor Antagonists
1.1 Catalysts: Scandium triflate Solvents: Acetone; rt		
		By: Eda, Masahiro; et al
Experimental Protocols		Journal of Medicinal Chemistry (2015), 58(12), 4918-4926.



31-260-CAS-18055031	Steps: 1	Synthesis and Biological Evaluation of Cyclopenta quinoline Derivatives as Nonsteroidal Glucocorticoid Receptor Antagonists
1.1 Catalysts: Scandium triflate Solvents: Acetone; rt		
		By: Eda, Masahiro; et al
Experimental Protocols		Journal of Medicinal Chemistry (2015), 58(12), 4918-4926.